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May-25-1097

MCA-6201 (Data Structure Using C)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Sections A, B, C and D. Section E is compulsory. All questions carry marks as indicated.

SECTION - A

1. Define and classify data structures. How are complexity and efficiency related to data structures? (12)

OR

2. (a) What is an ADT? Explain with an example of a List ADT. (8)
(b) Compare time complexities of arrays and linked lists for insertion, deletion, and search operations. (4)

SECTION - B

3. Explain various types of linked lists. Write a C program to reverse a singly linked list. (12)

OR

4. (a) Differentiate between Stack and Queue with examples. (6)
(b) Write a C function to implement a queue using arrays. (6)

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SECTION - C

5. Explain AVL Trees. Write rotation cases in AVL trees with diagrams. (12)

OR

6. (a) Explain the concept of threaded binary trees. How do they differ from binary search trees? (8)
(b) What is the difference between B-Tree and B+Tree? (4)

SECTION - D

7. Explain Merge Sort algorithm with an example. Derive its time complexity in best, average, and worst case. (12)

OR

8. (a) Differentiate between Quick Sort and Radix Sort. (6)
(b) Write a C program to perform selection sort on an array. (6)

SECTION - E (Compulsory)

9. Answer briefly.

- (a) Define Space Complexity.
(b) What is Sparse Matrix? Give an example.
(c) Differentiate between static and dynamic memory allocation.
(d) What is double-ended queue (Deque)?
(e) Explain Big-Omega notation with example.
(f) List two applications of graphs. (6×2=12)